

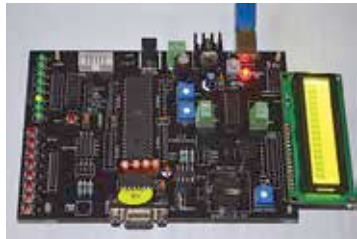
All these functions are carried out by the help of embedded systems. Why should you know about them? Well, normally users frolic in the joy that these technologies bring without ever understanding their intricacies. Don't you want to stay one step ahead of the rest? Also, as cited before, your home is probably full of them, and wouldn't it be a good idea to understand these little pixies of the electronics world?

Remember greeting cards? Actual ones, not those pathetic e-versions. Now remember the musical ones? The music starting to play when you opened the card was powered by a simple 4-bit embedded system. More complex 128-bit embedded systems are used in nuclear power plants, satellites, aircraft guidance systems because, well, these systems are more complex.

So what is an embedded system?

An embedded system is a microprocessor-based device which has an inbuilt operating system. This OS is designed to perform specific functions, that too under a time constraint. For example, take a standard (dumb) mobile phone. You don't just want to call on it.

You would like to be able to reduce the screen brightness, to be able to adjust the zoom of the camera, or you may want to use it as a calculator. To bring about such variations in its functionality, embedded processors come to the rescue. Each embedded processor will handle one aspect such as screen brightness, or FM radio, and so these



Li2_Board_LCD

and all the other embedded processors for each specific function will collectively be called an embedded system. Moreover, it is working under a time constraint as well – you don't want to call up a friend and have to wait for ten minutes to hear what they are saying. This is called working in 'Real time'. A situation like that where you have to wait for more time before talking is not going to cause a life-threatening catastrophe (at least we guess that it won't) and so this system is called a soft real-time system. But if you consider, say, the working of a pacemaker (a small device that's placed in the chest or abdomen to help control abnormal heart rhythms – you can find the working in the next chapter), it just cannot afford to have even the tiniest delays in execution, and so a system like this is called a hard real-time system.